IGCSE 0607 Practice Questions — E1.8

Standard form

Total Questions: 18 — **Total Marks:** 93 Marks

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1 **(a)** Write 0.00407 in standard form.

..... **[1]**

(b) Work out $(3 \times 10^4) \times (5 \times 10^6)$.
Give your answer in standard form.

..... **[2]**

2 **(a)** Write 450000 in standard form.

..... **[1]**

(b) Write 3.08×10^{-4} as an ordinary number.

..... **[1]**

(c) Work out.

$(2 \times 10^3) \times (4 \times 10^5)(0)$

Giveyouranswerinstandardform.

..... **[1]**

3 **(a)** Write 452000 in standard form.

..... **[1]**

(b) Write 6.07×10^{-3} as an ordinary number.

..... **[1]**

(c) Work out $(4 \times 10^4) \times (3 \times 10^5)$.
Give your answer in standard form.

..... **[2]**

4 **(a)** Write 0.00714 in standard form.

..... **[1]**

(b) Write 5.02×10^4 as an ordinary number.

..... **[1]**

(c) Work out the value of

$$(3 \times 10^5) \times (6 \times 10^4)$$

Give your answer in standard form.

..... **[2]**

5 Consider the following numerical values.

(a) Convert the following numbers as instructed.

(i) Write 0.00034 in standard form.

..... **[1]**

(ii) Write 5.2×10^4 as an ordinary number.

..... **[1]**

(b) Work out, giving your answer in standard form:

$(2 \times 10^3) \times (4 \times 10^5)$

..... **[2]**

6 **(a)** Write 0.000452 in standard form.

..... **[1]**

(b) Write 3.71×10^5 as an ordinary number.

..... **[1]**

(c) Calculate, giving your answer in standard form.

$(2 \times 10^4) \times (4 \times 10^3)$

..... **[2]**

7 The mass of the Earth is 5.97×10^{24} kg.
The mass of the Moon is 7.35×10^{22} kg.

(a) Calculate the difference between the mass of the Earth and the mass of the Moon.
Give your answer in standard form.

..... kg **[2]**

(b) The mass of the Sun is 1.99×10^{30} kg.
Find the value of k when

mass of the Sun = k ×mass of the Earth
Give your answer to 3 significant figures.

$k =$ ..... **[3]**

8 **(a)** Work out $(8.5 \times 10^6) - (4 \times 10^5)$.
Give your answer in standard form.

..... **[2]**

(b) It is given that $p = 2 \times 10^9$ and $q = 5 \times 10^4$.
Work out the value of $\sqrt{\frac{p}{q}}$.
Give your answer in standard form.

..... **[3]**

9 **(a)** Work out, giving your answer in standard form.

$$\frac{3.6 \times 10^{-4}}{8 \times 10^{-7}}$$

..... **[3]**

(b) Work out, giving your answer in standard form.

$$(5 \times 10^{8}) + (7 \times 10^{7})$$

..... **[2]**

(c) Given that $x = 2 \times 10^{m}$ and $y = 8 \times 10^{n}$, where m and n are integers.
Find an expression for x^2y in standard form, in terms of m and n.

..... **[2]**

10 **(a)** Work out the exact value of

$$\frac{5.4 \times 10^5 + 6 \times 10^4}{2 \times 10^{-3}}$$

Give your answer in standard form.

..... **[3]**

(b) Evaluate $\sqrt{2.5 \times 10^9}$.
Give your answer in standard form.

..... **[2]**

(c) Given that $p = 2 \times 10^x$ and $q = 8 \times 10^y$, find an expression for $p^2 \div q$.
Give your answer in standard form in terms of x and y.

..... **[3]**

11 $p = 5 \times 10^m$ and $q = 8 \times 10^n$, where m and n are integers.

(a) Find an expression, in terms of m and n, for $p \times q$.
Give your answer in standard form.

..... **[2]**

(b) Find an expression, in terms of m and n, for $\frac{p^2}{q}$.
Give your answer in standard form.

..... **[3]**

(c) Given that $p + q = 5.08 \times 10^m$, work out the exact value of $m - n$.

$m - n =$ ..... **[3]**

12 The distance from the Earth to the Sun is 1.5×10^8 km.

(a) Light travels at a speed of 3.0×10^5 km/s.
Calculate the time taken for light to travel from the Sun to the Earth.
Give your answer in seconds.

..... seconds **[2]**

(b) A spacecraft travels at a constant speed of 2.5×10^4 km/h.
Work out the time it takes the spacecraft to travel from the Earth to the Sun.
Give your answer in hours, in standard form.

..... hours **[2]**

13 The mass of Venus is 4.87×10^{24} kg.
The mass of the Sun is 1.99×10^{30} kg.

(a) The mass of Earth is 1.23 times the mass of Venus.
Calculate the mass of Earth.
Give your answer in standard form correct to 3 significant figures.

..... kg **[2]**

(b) Calculate how many times greater the mass of the Sun is than the mass of Venus.
Give your answer in standard form correct to 3 significant figures.

..... **[2]**

14 The mass of the Sun is 1.99×10^{30} kg.
The mass of Jupiter is 1.90×10^{27} kg.

(a) Calculate how many times greater the mass of the Sun is than the mass of Jupiter.
Give your answer in standard form correct to 3 significant figures.

..... **[3]**

(b) A comet has a mass of 4.5×10^{13} kg.
Calculate the mass of the comet in grams.
Give your answer in standard form.

..... g **[2]**

15 **(a)** Calculate, giving your answer in standard form.

$3.4 \times 10^7 + 2.1 \times 10^6$

..... **[2]**

(b) A computer can do a calculation in 4×10^{-9} seconds.
Work out how many of these calculations the computer can do in 5 minutes.
Give your answer in standard form.

..... **[3]**

16 The mass of the Earth is approximately 5.97×10^{24} kg.
The mass of the Moon is approximately 7.35×10^{22} kg.

(a) Calculate the total mass of the Earth and the Moon.
Give your answer in standard form.

..... kg **[2]**

(b) Calculate how many times greater the mass of the Earth is than the mass of the Moon.
Give your answer correct to the nearest integer.

..... **[2]**

(c) The mass of a planet is $p \times 10^{25}$ kg.
The mass of this planet is 300 times the mass of the Earth.
Work out the value of p, giving your answer correct to 3 significant figures.

$p =$ ..... **[2]**

17 **(a)** The mass of Earth is 5.97×10^{24} kg and the mass of the Moon is 7.35×10^{22} kg.
Calculate the difference between the mass of Earth and the mass of the Moon.
Give your answer in standard form.

..... kg **[2]**

(b) The distance from the Sun to Earth is 1.50×10^{8} km.
The speed of light is 3.00×10^{5} km/s.

Calculate the time it takes for light to travel from the Sun to Earth.
Give your answer in standard form.

..... s **[2]**

(c) The Earth is modelled as a sphere with radius $r = 6.37 \times 10^{6}$ m.
Calculate the surface area of the Earth.
Give your answer in standard form, correct to 3 significant figures.
The surface area, A, of a sphere with radius r is $A = 4\pi r^2$.

..... m^2 **[3]**

18 A large cylindrical water tank has a base radius of 4.5×10^2 cm and a height of 1.2×10^3 cm.

(a) Calculate the volume of the water tank in cm^3.
Give your answer in standard form correct to 4 significant figures.

..... cm^3 **[3]**

(b) The empty tank is being filled with water at a constant rate of 1.5×10^{-4} m^3 per second.
Note that $1\ m^3 = 1 \times 10^6\ cm^3$.

Calculate the time taken to fill the tank completely.
Give your answer in hours, written in standard form correct to 3 significant figures.

..... hours **[4]**